1. Run report — twist obstruction, run-20261026T1456Z

Observations only. The interpretation is in `../../REPORT.pdf`.

Command	`twist-obstruction full reports/run-20261026T1456Z` (release build)
Commit	`ff3bf9d485ebb27350542f0203a6bac090f1835a` (harness and frozen `PROTOCOL.md`)
Protocol sha256	`ed94554d0e65fd814590bcc617439efc85d3a59c51c979fae97722f1055ef358`
Seed	`0x7715720261026` (splitmix64)
Host	Claude Code cloud container, Linux 6.18.44, 4 vCPU, unisolated, single thread
Wall time	269.0 s (recorded in `run.json`; not a result)
Rows	4969: 15 census rows and 4954 decomposition targets. Every requested row completed; there were no failures.
Raw records	`rows.jsonl` (sha256 `d8cab334…c41189`), `summary.json` (sha256 `4dd346bb…249f97`), `stderr.log`

1.1. Census (exhaustive, n = 7, 11, 13; 5 curves each)

Every P ∈ E(F_q) was paired with every Q ∈ φ(E′(F_q)). Measured:

- 356,633,920 pairs in total.
- Q not 2-torsion: 356,530,036 sums, of which **0** are σ-fixed.
- Q ∈ {O, T}: 103,884 sums, of which all are σ-fixed.
- Count checks, **0 mismatches**:
  - |σ = −1 part| = #E′(F_q), with #E′ counted independently on the twist model;
  - φ(E′(F_q)) lies in the σ = −1 part;
  - #E + #E′ = 2q + 2.

1.2. Decompositions (curve model E_a and twist model E_(a+1), same x(R))

All values are measured counts from `summary.json`. "Off" means a solution with at least one coordinate whose σ-parity is opposite to R's. The interval is a 95 % Clopper–Pearson interval, derived by `summarize/`. "Heuristic" is the solution-weighted 1 − p_E^m (derived).

arm	n	m	targets	solutions	off obs	off pred	off fraction [95 %]	heuristic
A	11	2	1200	4836	0	0	0 [0, 0.0008]	0.737
A	13	2	1200	4415	0	0	0 [0, 0.0008]	0.765
A	17	2	800	3178	0	0	0 [0, 0.0012]	0.750
A	19	2	600	2339	0	0	0 [0, 0.0016]	0.754
A	11	3	300	6582	416	416	0.063 [0.057, 0.069]	0.871
A	13	3	300	11377	463	463	0.041 [0.037, 0.045]	0.893
A	17	3	200	4256	0	0	0 [0, 0.0009]	0.863
A	19	3	150	6389	0	0	0 [0, 0.0006]	0.874
C1	11–19	3	170	39861	18042	18042	—	—
C2	11–19	3	34	1098	95	95	—	—

Decision-rule counters, summed over every arm and both models. **All are 0.**

- violations (off-parity sub-sum ∉ {O, T});
- witness failures;
- targets where the observed off set ≠ the predicted off set;
- targets where the x-set differs between E_a and E_(a+1);
- point solutions missing from the explicit-polynomial roots;
- unexplained extra roots.

Explicit-polynomial cross-checks:

- S_3 on all 3800 m = 2 targets: 14,768 roots, equal to the point sets.
- S_4 = Res_X(S_3, S_3) on 105 m = 3 targets: 2910 roots, equal to the point sets, with no degenerate extras.

Every tally (pure, 2-torsion, off_O, off_T) is identical on the curve model and the twist model for every arm. The p_E measured over A targets is 0.46–0.51 by family.

Positive controls:

- C1 (x_R planted in V): the predicted set is non-empty on 170 of 170 targets, and observed = predicted on every one.
- C2 (planted T-pairs): 95 off_T solutions in total, all predicted.

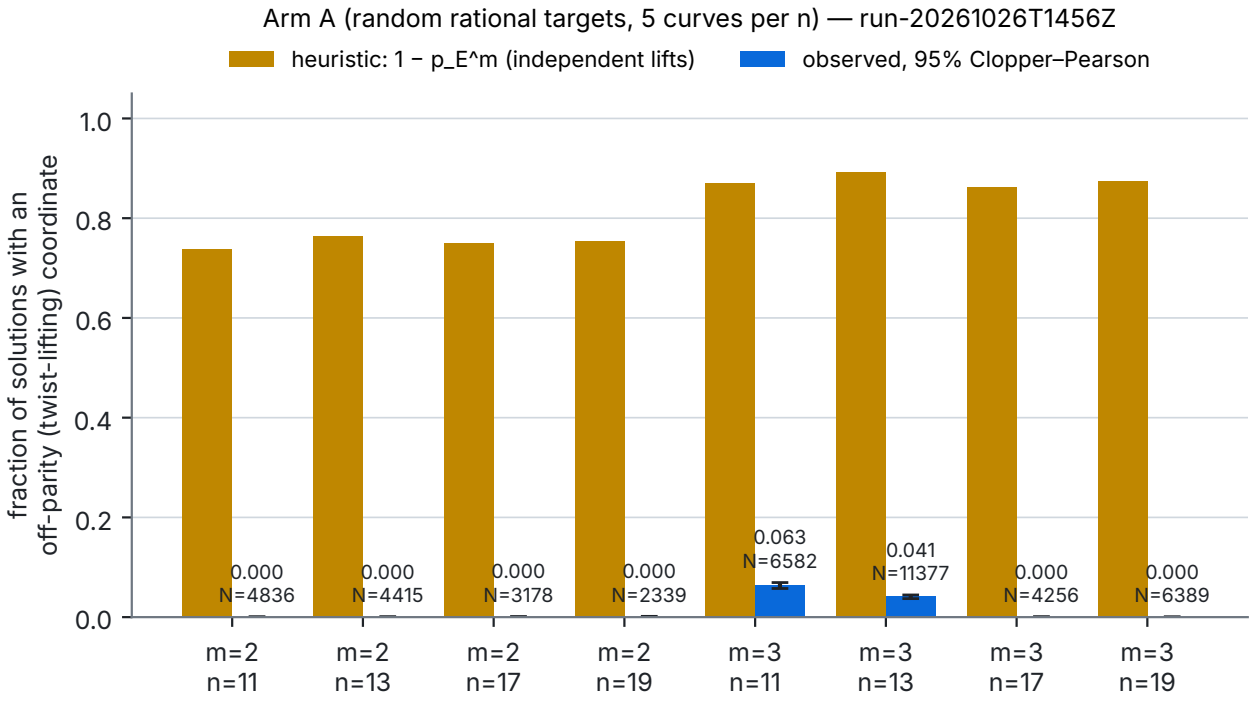


Figure 1: Arm A, run-20261026T1456Z. Observed fraction of solutions with an off-parity coordinate (95 % CP) beside the independent-lift heuristic. N = solutions per family.

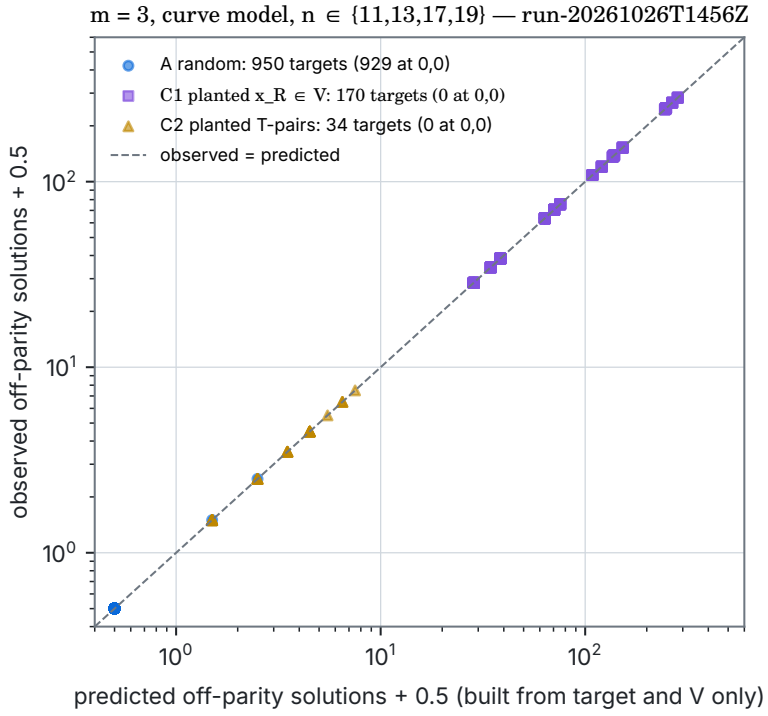
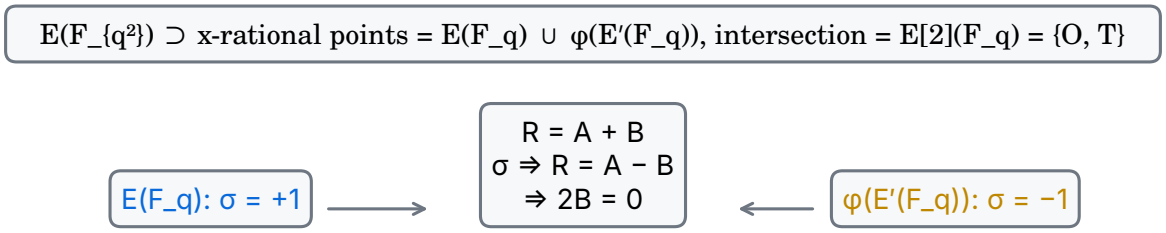


Figure 2: m = 3, curve model. Per target: predicted off-parity count (built from the target and V only) against the observed count. Arms A, C1 and C2; there are 1154 targets.



Summation polynomial S_{m+1}(x_1..x_m, x_R) = 0 with x_i ∈ V: off-parity sub-sum ∈ {O, T}.
m = 2: no mixed solution. m = 3: mixed solutions are exactly (x_R, x, x) and (x(R+T), x, x(P_x+T)).

Figure 3: Diagram of the eigenspace decomposition the run checks.